

KARTIK RAUT

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AI/ML Engineer · LLM & GenAI Specialist · RAG & Agentic Systems

SUMMARY

AI/ML Engineer with production-grade experience building LLM systems, RAG pipelines, agentic AI workflows, and RL-based autonomous agents. Shipped a real-time voice AI product achieving 1–2s latency and ~40% inference cost reduction; published a GRPO RL multi-agent system on HuggingFace Spaces. Proficient in the full MLOps lifecycle: data ingestion → model training → vector indexing → FastAPI/Docker deployment. Hackathon Finalist — Meta × Scaler PyTorch (31,000+ registrants) and Claude Solve-A-Thon IIIT Nagpur (Top 50/650+).

TECHNICAL SKILLS

Languages & Databases: Python, SQL, JavaScript, React, ChromaDB, FAISS, PostgreSQL, AWS (Basic)

Frameworks: PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, HuggingFace Transformers, XGBoost, LightGBM, Pandas, NumPy

Dev Tools: FastAPI, Flask, Docker, Git/GitHub, REST APIs, Streamlit, Plotly, Matplotlib, Seaborn

AI Specializations: RAG, Prompt Engineering, Agentic Pipelines, Finetuning (TRL/LoRA), Embedding Generation, Groq API, Whisper STT/TTS, Context Window Management, GRPO RL, Q-learning, U-Net, LSTM

MLOps & LLMops: Vector DB Design, Inference Optimization, CI/CD for ML, LLMops, Intent Routing, Docker Deployment

EDUCATION

GH Rasoni University, Amravati

Bachelor of Technology in Artificial Intelligence

Coursework: Machine Learning · Deep Learning · NLP · Statistics & Probability · Data Structures & Algorithms · DBMS · Computer Networks

Aug 2022 – May 2026

CGPA: 7.5 / 10.0

PROFESSIONAL EXPERIENCE

AI/ML Engineer Intern · AbCoEd Tech Pvt. Ltd., Nagpur

Dec 2025 – May 2026

- InfinoAI Voice AI (Agentic Product):** Engineered a production real-time voice assistant integrating Groq LLaMA 3.3 70B, Whisper STT/TTS, and an agentic intent-routing pipeline; achieved 1–2s end-to-end latency with consistent multi-turn accuracy across diverse user intents.
- ~40% Inference Cost Reduction (LLMops):** Architected a 7-category intent routing system that eliminated unnecessary LLM calls, reducing inference costs by ~40% and improving response latency — a measurable LLMops optimization that directly lowered operating expenditure.
- Production Deployment (Docker + Flask + React):** Implemented full voice pipeline (mic → STT → intent detection → execution → TTS) and deployed via Dockerized Flask backend with React dashboard; shipped production-ready demo within tight project timelines.
- Prompt Engineering & Context Management:** Designed structured prompt templates and context window strategies enabling reliable output format consistency and reduced hallucinations across 7 distinct task categories.

PROJECTS

CloudEdge — Multi-Agent LLM Simulator for Cloud Crisis Management

2026

[HuggingFace Spaces](#) | PyTorch · TRL · LLaMA 3.3 70B · GRPO · Docker

- GRPO RL Fine-tuning Pipeline:** Built a GRPO reinforcement learning system (30 episodes × 512 steps) training LLM agents for autonomous cloud infrastructure decisions; improved reward +48% (4.6 → 6.8) and reduced policy entropy from 0.50 → 0.02, demonstrating end-to-end fine-tuning pipeline mastery.
- Agentic Multi-Agent Architecture:** Designed a 3-agent negotiation system (resource, cost, sustainability agents) across 7 crisis scenarios with tool use and inter-agent communication; published live on HuggingFace Spaces via Docker.

UrbanSense — Full-Stack Geospatial AI Platform for Urban Risk Analysis

2026

[GitHub](#) | FastAPI · React · PostGIS · Docker · PyTorch · XGBoost

- 5-Module ML System at Scale:** Built a geospatial AI platform covering 200+ sq. km with U-Net segmentation, XGBoost/LightGBM risk models, LSTM forecasting, and graph analysis on 50,000+ data points; delivered 20+ REST APIs with interactive visualizations.
- 4-Metric Evaluation Framework:** Established systematic model evaluation using IoU, F1, ROC-AUC, and RMSE — aligned with production ML standards for bias detection, accuracy, and deployment readiness.

Documentation Analyzer — Intelligent RAG Crawling Agent

2025

[GitHub](#) | LangChain · ChromaDB · FAISS · Groq API · Streamlit

- RAG Pipeline with Vector Indexing:** Engineered a BFS crawler with deduplication and chunking; indexed 500+ text chunks across 10+ PDFs in ChromaDB using embedding generation and FAISS vector indexing; integrated Groq LLaMA 3.1 8B for sub-second citation-based answers.

ACHIEVEMENTS & CERTIFICATIONS

- Finalist** — Meta × Scaler PyTorch OpenEnv Hackathon, Bangalore: Top teams from 31,000+ registrations; 48-hour on-site AI hackathon — validated rapid prototyping and production ML deployment under pressure.
- Finalist** — Claude Solve-A-Thon, IIIT Nagpur: Top 50 out of 650+ participants in 18-hour on-site AI hackathon — demonstrated LLM engineering, problem decomposition, and delivery speed.
- Certifications:** Machine Learning Specialization — Andrew Ng, Coursera | Python Essentials — Scaler